Charitha Sri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.25
Narayana Junior College, Kothapet, Telangana	Class XII (TSBIE)	2024	9.72%
New Vision High School, Khammam, Telangana	Class X (SSC)	2022	10.00%

SKILLS

- **Languages** : Python, C++, SQL
- **Frameworks** : PyTorch, scikit-learn, SciPy, NumPy, pandas, Matplotlib, dice-ml
- **Tools** : Git, GitHub, Linux

RELEVANT COURSES

Matrix Theory, Optimization - I, Programming for AI, Probability and Random Variables, Foundations of ML, Data Structures and Applications, Deep Learning, Optimization - II, Algorithms, AI for Humanity

PROJECTS

Trustworthy PMOS prediction:

- Built a trustworthy AI pipeline for PMOS (formerly PCOS) prediction integrating fairness auditing, calibration, conformal prediction, SHAP explainability, counterfactual explanations, and decision curve analysis
- Trained **XGBoost** on a Kaggle dataset of 541 patients with staged feature engineering; cheap clinical features caught the same number of PMOS cases as expensive lab tests (83% recall), validating cost-effective screening; full model achieved 92.7% accuracy and *0.951 AUC*
- Applied **conformal prediction** to defer uncertain cases to clinicians; recall improved from 83.3% to 92.9% on action-ed cases
- Used **SHAP** for global and local explanations, generated **counterfactual explanations** identifying clinically actionable changes

W-DRO Heatwave Forecast:

- Built an LSTM-based time-series forecasting model to predict heatwaves 7 days ahead
- Applied **Wasserstein** Distributionally Robust Optimization (W-DRO) - formulated as a min-max optimization problem - to make the model robust under distribution shifts, using *projected gradient ascent* to solve the inner maximization
- Demonstrated that w-dro mitigates performance degradation when future climate patterns diverge from training data, ensuring more reliable predictions compared to standard LSTM baselines.

Fair Income Project:

- Built an XGBoost classifier and audited its predictions using Equalized Odds - male predictions were systematically over-estimated while female predictions were under-estimated
- Evaluated cluster based SMOTE (FAPFID) to balance protected groups within features; fairness improved marginally, showed that bias is embedded in correlated features (proxy bias) rather than group imbalance alone
- Implemented **post processing threshold tuning,** reducing Equalized Odds to 0.072 while preserving accuracy (83.3%) and recall (84%)

Deherding Recommendation System: (*Group Project*)

- Built a collaborative recommendation system that mitigates herding effects caused by popularity bias and conformity bias
- Implemented **Smooth xQuAD,** Calibrated Popularity with wasserstein distance, and MACR (Model-Agnostic-Counterfactual-Reasoning) to reduce popularity bias through re-ranking and causal inference
- Achieved highest **long-tail coverage** and **Nash Social Welfare** with Smooth xQuAD, ensuring equitable niche discovery across users.

POSITIONS OF RESPONSIBILITY

Literary Society Core Member | IIT Hyderabad (May 2025 - April 2026)

- Published poetry in Lexicon and contributed to content strategy and promotional design

EXTRACURRICULAR

- Creative and technical writer on Medium